

Battery Analysis Tool · Quick Start Guide

For BOEDT Battery Analysis Tool

This tool analyzes **Excel data** exported from the NEWARE battery test system and automatically generates box plots, Excel result sheets, Word reports, and CSV summaries.

Compared to previous versions, the `setting.ini` file has been removed. All configuration is now done within the app (**Tools** → **Configuration**). A batch conversion tool, `ndaxConverter`, ships alongside this release and is available at the same download page as the main software.

1. Fastest Way to Get Started: Three Steps

Step	Action	What the app does automatically
1	Select Test Profile (Browse to an XML test profile file)	See Tips 1 below
2	Click the green Run	Validation → calculation → plotting → report generation, fully automatic
3	When done, click "Open Report" / "Open Path"	One-click access to your results

- In most cases you only need to **select one XML file and click Run once** — the rest is fully automatic.

 **To add new options (e.g. a new battery type or manufacturer), open Configuration and add a new entry or row there.**

2. Time-Saving Tips (Highlights)

Tips 1. Select One XML, and All Paths Are Auto-Filled

After selecting a **Test Profile (XML)**, the app automatically:

- Validates the XML (file name, content; encoding auto-compatible with UTF-8/GBK)
- **Sets the input path** to the sibling `2_xlsx` folder
- **Sets the output path** to the sibling `3_analysis_results` folder (asks whether to create it if it doesn't exist)
- **Detects temperature automatically**: `Freezer` in the file name → Freezer temperature; otherwise → Room temperature
- Calculates and fills in the version number
- If you are a new team member, just type your name directly into the name field.

💡 If the output folder doesn't exist, a dialog will ask whether to create it — click "Yes" and you won't need to create folders manually.

Tips 2. Select a Directory, and the Form Is Auto-Filled

After setting the input path, the app scans all `.xlsx` files in the directory **in a background thread** (the UI stays responsive) and auto-fills:

- **Samples Qty** = number of Excel files in the directory
- Parsed from the **first file name**:
 - Construction Method
 - Specification Type (automatically matches the longest one)
 - Specification Method
 - Manufacturer
 - **Batch/Date Code** (supports `DC2604`, `DC(2604)`, and `DC (2604)` formats)
 - **Pulse Current** (extracted from patterns like `(15-6mA)`)
 - **CC Current** (extracted from patterns like `mA, xxmA`)

💡 Put the specification/batch/current info clearly in your file names — after selecting the directory, the form is almost fully filled in, and you just confirm.

Tips 3. Capacity and Temperature Are Auto-Determined

After selecting "Specification Type + Specification Method", the app matches the **Rules** in the configuration and auto-fills:

- **Datasheet Nominal Capacity, Calculation Nominal Capacity, Required Useable Capacity** (= calculation capacity × required percentage, e.g. 80%)

- Automatically matches the **Battery Type** (e.g. selecting CR2450 → Coin Cell)
- Automatically determines the **temperature** based on required useable capacity:
 - 280 → switches to **Freezer -20°C**
 - 380 → switches to **Room temperature**

💡 If the freezer temperature is set to 0°C when you click Run, a confirmation dialog appears to prevent mistakes.

Tips 4. Zero Configuration on First Launch

- On first run, a default configuration is created automatically at `%APPDATA%\battery-analysis\config.json`
 - Battery types, specifications, manufacturers, specification methods, pulse currents, cut-off voltages, tester locations, testers, equipment models and firmware versions are **all pre-set**
 - Old configurations are migrated automatically, with automatic integrity validation
 - Missing/empty configuration values fall back to **built-in defaults**, so the UI never goes blank due to a broken config
 - Loaded automatically on every launch — no need to re-set up
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Tips 5. Test Information Is Auto-Saved and Auto-Loaded

- Content in the **Test Information table** (equipment, software versions, testers, ambient temperature/humidity, etc.) is saved to the configuration automatically when you click Run
 - It is auto-loaded next time you open the app
 - Drop-downs (Tester Location, Tested By, Reported By) load automatically from the configuration
 - **Selecting a "Tester Location" auto-fills the whole equipment table:** test equipment, BTS server/client/analysis software versions, middle machine and test unit model/hardware version/serial number/firmware — all pre-set in the configuration per location, filled in with one click
 - After editing the configuration dialog and closing it, your previously selected Battery Type, Manufacturer, Tester Location, Tested By, etc. are **restored automatically** — nothing is lost
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Tips 6. One-Click Access After Analysis Completes

- When analysis finishes, a dialog offers **"Open Report"** / **"Open Path"** / **"OK"**
 - Automatically matches the current version of the Word report; if not found, the **latest** one is opened
 - Report files are named automatically:
`Manufacturer_Specification_DCBatch_TDDate_vVersion.docx`
 - **Word template is chosen automatically** based on the number of current levels (≤ 4 or >4)
 - Output directory is created automatically (`Date_vVersion` subfolder) and contains: box plots (PNG/SVG), Excel result sheets, Word report, CSV summary
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Tips 7. Language and Theme Are Remembered; Window Auto-Adjusts

- **Language:** auto-detects the system language on first launch; once you switch manually it's remembered for next launch
 - **Window:** centered and sized to fit the screen automatically on every launch (scroll bars appear on small screens; the window can be freely resized/maximized)
 - **Theme / Font size:** set in Preferences and remembered automatically
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Tips 8. Real-Time Validation — You'll Know Immediately When Something's Wrong

- When a required field is empty, a path doesn't exist, or the version format is invalid, the corresponding input box turns **red** and the status bar shows a hint
 - When validation passes, the status bar shows `status:ok` and the Run button is ready
 - Problem files are listed in a dialog with detailed reasons for easy troubleshooting
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3. Menus

- **Tools** → **Configuration:** manage battery config, test config, and equipment info (add/remove specifications, manufacturers, testers, etc.)
 - **Preferences:** language, theme, font size, auto-save, exit confirmation, etc.
 - **Help** → **User Manual:** view this guide
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4. FAQ

Problem	Solution
"Input path has no data"	Make sure <code>.xlsx</code> files exist in the <code>2_xlsx</code> folder (files starting with <code>~\$</code> are ignored as temp files)
"Output directory does not exist"	Click "Yes" in the dialog to create it automatically, or pick an existing directory manually
Version number is wrong	The version is determined automatically by the data content — don't edit it manually; check that the input/output paths are correct
UI appears incomplete	The window can be freely resized/maximized; a scroll bar appears automatically on small screens

During repeated testing and use, I made many user-friendly fixes and improvements based on commonly used software. However, due to insufficient testing time and samples, there are inevitably still many bugs. If you have any questions, please let me know and I will fix them.